

How the Climate Academy AI Assistant Works

A plain-language guide for colleagues (non-technical readers)

Climate Academy Student Book · Matthew Pye (2025)

What it is

The Climate Academy Assistant is a chat tool that answers questions using the Climate Academy Student Book. Think of it as a patient study partner who has the textbook open and tries to answer only from that book - not from random websites or general "internet knowledge."

You type a question in normal language (English, Hindi, French, or others). The assistant replies in the same language you used, with short, clear answers and references to book sections (for example Section 1.2.3).

What it is not

Myth	Reality
"It knows everything about climate."	It only uses text prepared from this book.
"It is always right."	It can misread or combine passages; treat it as a helper, not a teacher.
"It browses the web live."	It searches a pre-built index of book passages on the server.
"It remembers me forever."	It remembers only the current chat (recent messages), not your name or other info.

The big picture (three steps)

YOU		BOOK (prepared once)		ANSWER WRITER
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Ask a question in the chat	-->	Passages indexed by meaning (searchable "smart index")	-->	Writes reply using ONLY those passages + section citations
Read the answer in the chat	<--			

1. Prepare the book (once) - The book is split into many small passages and indexed so the system can find "the parts most like your question."
 2. Find passages (every question) - Your question is matched to the top few passages from the book.
 3. Write the answer (every question) - A language model drafts a reply using only those passages, with rules like "do not invent facts" and "cite section numbers."
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Step 1 - Preparing the book (before anyone chats)

Like building a very smart index at the back of a textbook:

```
Student Book (HTML)
|
v
Split into sections & paragraphs (section numbers: Section 1.2.3 ...)
|
v
Cut into overlapping passages (a few hundred words each)
|
v
"Fingerprints" for meaning (so similar ideas match, not just keywords)
|
v
Search index stored on disk
```

Analogy: A librarian cuts the book into labelled index cards - each tagged with "Section which section" - and files them in a cabinet sorted by meaning, not only alphabetically.

Step 2 - When someone asks a question

Colleague	Chat screen	Passage finder	Book index
-- question ----->			
	-- find similar --->		
		--- search ----->	
		<-- top 5 passages	
	<-- passages -----		
	--- passages + rules + history ----->	AI writer	
<-- answer -----			

Semantic search means the system looks for passages about the same idea as your question, not only the exact same words.

Example: a question about "why Earth is warming" can still find a passage about the "greenhouse effect" even if you never used those words.

Step 3 - How the answer is written

The answer writer is a large language model (in our setup: Llama 3.3 via Groq). It does not freely invent climate facts. It is given strict instructions, including:

- ? Answer only from the passages retrieved in step 2.
- ? Reply in the same language as the user.
- ? If the book does not cover the topic, say: "I could not find information about that in the Climate Academy book."
- ? Cite section numbers (e.g. Section 1.2) when using information from a passage.
- ? Stay concise and encouraging; use bullet points when listing several items.

Analogy: A teaching assistant who may only quote from the highlighted pages you handed them - not from memory of other textbooks.

Example question and answer

Colleague asks:

What is the greenhouse effect?

Behind the scenes: The index returns passages from a section such as "The greenhouse effect", including: certain gases trap heat; sunlight warms the surface; the surface emits infrared radiation; greenhouse gases absorb and re-emit heat.

The assistant might reply (illustrative):

The greenhouse effect is the process by which certain gases in the atmosphere trap heat, keeping Earth warmer than it would otherwise be. Without it, the planet would be much colder.

>

In short:

- Sunlight (shortwave) warms the surface.*
- The surface emits infrared radiation.*
- Greenhouse gases absorb and re-emit part of that infrared, so heat is retained.*

>

Key gases in the book include carbon dioxide (CO₂) and methane (CH₄), with water vapour as a fast feedback (Section ... - The greenhouse effect).

(Exact Section numbers come from the live book; the app inserts real section labels from the passages it found.)

Example - when the book does not have the answer

Question:

What will global temperatures be in the year 2100?

Expected behaviour:

I could not find information about that in the Climate Academy book.

The assistant is not meant to guess future temperatures from general knowledge.

Languages and follow-up questions

? Write in French ? the assistant is instructed to answer in French, using the same English book passages

as sources.

? Recent chat messages (about the last ten turns) are included so you can ask "Can you explain that more simply?"

? The sidebar can suggest starter questions (causes of climate change, renewable energy, etc.).

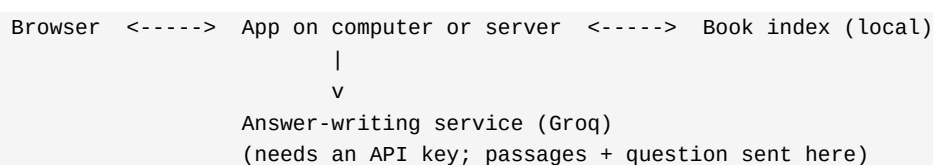
Where colleagues use it

Depending on deployment:

? On a laptop (demo / testing): Someone runs the app; testers open a link in the browser.

? On a small server (planned): A stable web address - colleagues only need a browser.

The logic is the same: browser ? question ? search book index ? write answer ? show in chat.



Privacy note: Questions and retrieved passages are sent to the answer-writing service (Groq) to produce the reply. The book index stays on your machine or server. Check your organisation's policy before using sensitive topics.

Quality and limitations

Strength	Limitation
Grounded in your official book	Weak if the topic is not in the book
Section citations (Section) help verification	Humans should still confirm citations
Works in multiple languages	Quality may vary by language
Fast, friendly explanations	Not a substitute for teaching, exams, or policy

Good uses: revision, exploring book content, drafting explanations students can check against the book.

Poor uses: exam answers without verification, legal or policy decisions, emergency advice.

One-sentence summary

The Climate Academy Assistant searches pre-indexed passages from the Student Book, then an AI writes a short answer only from those passages - in your language - with section references - and says clearly when the book does not cover your question.

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